

Statistics & Data Analysis — Master Cheat Sheet

Built to teach from zero: every topic → plain-English meaning → recipe → worked example. Prof. A. Giloni.

Summer 2026
Classes 01–04 + HW
Sample Midterm

How to read this: Blue box = formula. **HOW** green = method/example. **TRAP** red = mistakes that lose points. Purple = **step-by-step recipe**. Highlighted = **final answer**.
§18 is a "which formula?" finder; §19 is every Sample-Midterm problem in one table.

SYMBOLS — MEMORIZE THIS ROW $\bar{x}$ =sample mean · μ =population/true mean · s =sample SD · σ =population SD · σ^2 =variance · Σ =add up · n =sample size · π =success chance (binomial) · λ =avg count (Poisson) · ρ =correlation · $P(A)$ =chance of A · $|$ ="given" · $\cap$ ="and/both" · $\cup$ ="or/either" · A' ="not A" · $C(n,k)={}_nC_k$ ="n choose k".

§1 Types of Data & the Big Vocabulary

Class 1

Sort every variable 4 ways

- Quantitative** = numbers you can do math on (price, temp). **Qualitative/categorical** = labels (country, brand).
- Discrete** = counted, no in-between (# cars). **Continuous** = measured, in-betweens exist (gallons, time).
- Nominal** = categories with no order (Ford, Nissan). **Ordinal** = categories with an order (disagree → agree).
- Univariate** = 1 column. **Multivariate** = 2+ columns. **Time series** = order in time matters; **cross-sectional** = one snapshot.

Population vs Sample (the mental model)

- Population** = ALL items (fixed, usually can't collect). **Sample** = a subset you actually measure (random).
- Parameter** = a number describing the population (μ, σ) → **FIXED**. **Statistic** = a number from the sample ($\bar{x}, s$) → **RANDOM** (changes each sample).
- Random sample** = everyone has an equal chance to be picked → the sample resembles the population.

WHY IT MATTERS Every formula later comes in a **population** version ($\mu, \sigma, +N$) and a **sample** version ($\bar{x}, s, +n-1$). Using the wrong one is the #1 silly error.

§2 Describing the Center

Class 1

MEAN (AVERAGE)

$$\bar{x} = (x_1 + x_2 + \dots + x_n) / n = \Sigma x_i / n \quad (\text{population: } \mu = \Sigma x_i / N)$$

Median = middle of the sorted list. n **odd** → the true middle; n **even** → average of the two middle values. **Mode** = most frequent value. **Trimmed mean** = drop top & bottom x 's, then average (kills outliers).

MEAN VS MEDIAN The **mean gets dragged** by outliers & skew; the **median doesn't**. Right-skew (long right tail) → mean > median. So median is the "typical" value when data is lopsided.

EXAMPLE — MEAN & MEDIAN (N=9)

Sorted: 86.24, 88.77, 88.95, 94.62, **95.42**, 95.60, 95.64, 97.18, 99.80. Median = 5th = **95.42**. Sum=842.22 → mean=842.22/9=**93.58**. Mean < median → slight left pull.

§3 Describing the Spread (Variability)

Class 1

SAMPLE VARIANCE — DIVIDE BY N-1

$$s^2 = \Sigma (x_i - \bar{x})^2 / (n - 1) \quad s = \sqrt{s^2}$$

POPULATION VARIANCE — DIVIDE BY N

$$\sigma^2 = \Sigma (x_i - \mu)^2 / N \quad \text{shortcut } (\Sigma x_i^2 - N\mu^2) / N \quad \sigma = \sqrt{\sigma^2}$$

RECIPE · S^2 BY HAND

- Find $\bar{x}$.
- Each value: subtract $\bar{x}$, then square.
- Add all the squares.
- Divide by $n-1$. That's s^2 . Then $\sqrt{}$ for s .

GOTCHA SAMPLE ÷ $(n-1)$; POPULATION ÷ N . SD is just $\sqrt{\text{variance}}$ — its units match the data; variance units are **squared**. Variance/SD are never negative.

EXAMPLE — VARIANCE & SD

Data 2,4,6,8 (sample, $n=4$). $\bar{x}=5$. Deviations -3,-1,1,3 → squares 9,1,1,9 → $\Sigma=20$. $s^2=20/(4-1)=$ **6.67**, $s=\sqrt{6.67}=$ **2.58**.

Z-SCORE — STANDARDIZE A VALUE

$$z = (x - \bar{x}) / s = (x - \mu) / \sigma \rightarrow \text{"how many SDs from the mean"}$$

$z > 0$ above avg, $z < 0$ below. **Outlier rule:** $|z| \geq 3$ → outlier; $2 \leq |z| < 3$ → borderline.

EMPIRICAL RULE (68-95-99.7) Only if bell-shaped & symmetric: $\approx 68\%$ within $\mu \pm \sigma$ · $\approx 95\%$ within $\mu \pm 2\sigma$ · $\approx 99.7\%$ within $\mu \pm 3\sigma$.

Ex: $x=90, \mu=75, \sigma=10 \rightarrow z=(90-75)/10 =$ **+1.5** (1.5 SDs above avg).

§4 Quartiles, IQR, Box Plots & Outliers

Class 1

POSITIONS IN THE SORTED LIST

Q1 at $(n+1)/4$ · Q2 = median at $(n+1)/2$ · Q3 at $3(n+1)/4$

SPREAD MEASURES

Range = max – min (outlier-sensitive) · IQR = Q3 – Q1 (outlier-resistant)

BOX-PLOT FENCES (THE OUTLIER TEST)

Inner: $Q1 - 1.5 \cdot IQR$ & $Q3 + 1.5 \cdot IQR$ · Outer: $\pm 3 \cdot IQR$

EXAMPLE — BOX PLOT {30,40,70,81,90,118,123}

$Q1=70$, median=81, $Q3=90 \rightarrow IQR=20$. Fences: $70 - 1.5(20)=$ **40** & $90 + 1.5(20)=$ **120**. Outside → **outliers 30 & 123**; whiskers reach 40 & 118.

HISTOGRAMS Shape shows **skew** (which tail is long), **outliers** (isolated bars), **bimodal** (two humps). A **log transform** pulls a right-skewed data set toward symmetric. Bar chart = qualitative (gaps); histogram = quantitative (no gaps).

Past inner fence = **outlier**; past outer = **serious outlier**. Box spans Q1 → Q3, line = median, whiskers reach the farthest points still inside the fences; dots beyond = outliers.

§5 Probability — All 7 Rules, Labeled

Class 2

Setup words: Experiment=chance action · Outcome=one result · Sample space S=list of ALL outcomes · Event A=a group of outcomes · Mutually exclusive=can't both happen.

1 Range: $0 \leq P(A) \leq 1$
(0=impossible, 1=sure)

2 Total: $\Sigma P(\text{all simple events}) = 1$

3 Complement: $P(A^c) = 1 - P(A)$ ← best for "at least one"

4 Addition (OR): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

5 Mutually excl.: then $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$

6 Multiplication (AND): $P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$

7 Independence: $P(A \cap B) = P(A) \cdot P(B)$
⇔ $P(A|B) = P(A)$

§6 Conditional Probability & Independence

Class 2 / Handout

CONDITIONAL — "A GIVEN B"
 $P(A|B) = P(A \cap B) / P(B)$

You already KNOW B happened → shrink the whole world down to just B, then ask about A.

Independent vs dependent: independent = B happening does not change A's chance, so $P(A|B) = P(A)$ and $P(A \cap B) = P(A) \cdot P(B)$. Dependent = it does change it → you must use $P(A|B)$.
Test it: compute $P(A) \cdot P(B)$; if it equals the given $P(A \cap B)$ they're independent, else not.

EXAMPLE 1 — TWO COINS
S={HH, HT, TH, TT}, each ¼. A="at least one head"={HH, HT, TH}; B="at least one tail"={HT, TH, TT}. $A \cap B$ ={HT, TH}. $P(A \cap B) = 2/4 = 1/2$; $P(A \cup B) = P(\text{all four}) = 1$.

EXAMPLE 2 — IS IT INDEPENDENT?
 $P(A) = .4$, $P(B) = .2$. If independent: $P(A \cap B) = .4 \cdot .2 = .08$, $P(A \cup B) = .4 + .2 - .08 = .52$. If instead $P(A|B) = .6 \neq .4$ → **NOT independent**; then $P(A \cap B) = .6 \cdot .2 = .12$, $P(B|A) = .12 / .4 = .30$.

EXAMPLE 3 — ROLL TWO DICE (S HAS 36 EQUALLY-LIKELY OUTCOMES)
A="sum is 5", B="first die is 1 or 2", C="sum is 7". Count outcomes over 36:
 $P(A) = 4/36 = 1/9$ · $P(B) = 12/36 = 1/3$ · $P(C) = 6/36 = 1/6$ · $P(A \cap B) = 2/36 = 1/18$ ·
 $P(A \cup B) = 1/9 + 1/3 - 1/18 = 7/18$ · $P(B|A) = P(A \cap B) / P(A) = (1/18) / (1/9) = 1/2$ ·
 $P(C \cup B) = 1/6 + 1/3 - 1/18 = 4/9$ · $P(B|C) = (1/18) / (1/6) = 1/3$.

§7 Trees, Total Probability & Bayes

Class 2

TOTAL PROBABILITY — ALL PATHS TO AN EVENT
 $P(A) = P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)$

BAYES — FLIP $P(A|B)$ INTO $P(B|A)$
 $P(B|A) = P(A \cap B) / P(A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)}$

RECIPE · REVERSE THE CONDITION (BAYES)
1. **Bottom** = P(evidence) via total probability — add EVERY path that gives the evidence.
2. **Top** = the ONE path you're asked about.
3. Divide: top ÷ all-paths.

EXAMPLE — JAR GAME (SAMPLE EXAM Q1 & Q2)
Die: 5–6 → Jar 1 (15R, 5G); 1–4 → Jar 2 (8R, 12G). $P(J1) = 2/6 = .333$, $P(J2) = 4/6 = .667$;
 $P(G|J1) = 5/20 = .25$, $P(G|J2) = 12/20 = .60$.
P(Green) = $.25(.333) + .60(.667) = .0833 + .40 = .4833$ · **P(rolled 5,6 | Green)** = $.0833 / .4833 = .1724$ ·

EXAMPLE — MEDICAL TEST (RARE DISEASE)
 $P(\text{Dis}) = .01$, $P(\text{Pos}|\text{Dis}) = .98$, $P(\text{Pos}|\text{noDis}) = .01$. $P(\text{Dis}|\text{Pos}) = (.98 \cdot .01) / [(.98 \cdot .01) + (.01 \cdot .99)] = .0098 / .0197 = .497$ · Even a "good" test is a coin-flip when the disease is rare.

EXAMPLE — COIN PICKS A JAR (HW2)
Jar A 7R/3G, Jar B 15R/30G; fair coin (H → A, T → B). $P(\text{red}) = .5(.7) + .5(1/3) = .517$ ·
 $P(\text{tails}|\text{green}) = [.5(2/3)] / [.5(.3) + .5(2/3)] = .690$ ·

§8 Contingency (Two-Way) Tables

Class 2 / HW

	USA	Europe	Latin Am.	Asia	Other	Total
Male	22	5	4	6	3	40
Female	13	3	1	3	2	22
Total	35	8	5	9	5	62

Read straight off the table

- $P(\text{European}) = 8/62 = .129$ (column total ÷ grand total)
- $P(\text{European AND Female}) = 3/62 = .048$ (the one cell ÷ grand total)
- $P(\text{Male OR non-USA}) = (40 + 27 - 18) / 62 = 49 / 62 = .79$

Conditionals shrink the world

- $P(\text{Asian} | \text{Female}) = 3/22 = .136$ (stay in Female row ÷ its total 22)
- $P(\text{Male} | \text{not European}) = (40 - 5) / (62 - 8) = 35 / 54 = .648$

AND VS GIVEN "AND" divides by the **grand total (62)**. "GIVEN" divides by that **row/column subtotal**. Mixing these up is the #1 table mistake.

EXAMPLE — "NEITHER" & "ONLY ONE" (HW1) + DE MORGAN
29% hold an MBA (A), 24% a business degree (B), 8% both. $P(A \cup B) = .29 + .24 - .08 = .45$.
P(neither) = $P(A^c \cap B^c) = 1 - P(A \cup B) = .55$ · **P(MBA but not business)** = $P(A)$ - $P(A \cap B) = .29 - .08 = .21$ ·
Memorize: "neither" = $1 - P(A \cup B)$; "A but not B" = $P(A) - P(A \cap B)$. Turns word problems into arithmetic.

§9 Counting: Factorials, Permutations, Combinations

Class 2

FACTORIAL = WAYS TO LINE UP N ITEMS

$$n! = n \cdot (n-1) \cdot \dots \cdot 1 \cdot 0! = 1 \cdot 4! = 24$$

PERMUTATION — ORDER MATTERS (RANKS, TITLES)

$${}_nP_r = n! / (n-r)!$$

COMBINATION — ORDER DOESN'T (A GROUP)

$${}_nC_r = C(n,r) = n! / [r!(n-r)!]$$

RECIPE · DON'T COMPUTE HUGE FACTORIALS

1. In $10!/7!$, the smaller factorial **cancels** the tail: $10!/7! = 10 \cdot 9 \cdot 8 = 720$.
2. For ${}_nC_r$, expand r terms on top, $r!$ on bottom, cancel. Only multiply down to what's left.

P OR C? "Rank / title / order / sequence" → Permutation. "Choose / committee / group / hand" → Combination (= permutation ÷ $r!$). Probability by counting = **favorable ÷ total**, using the SAME tool top & bottom.

EXAMPLES

Pres/VP/Treas from 10 (order): ${}_{10}P_3 = 10 \cdot 9 \cdot 8 = 720$. Committee of 3 from 8: ${}_8C_3 = 56$. Pick 2 of 5 firms, P(exactly A&B) = $1/{}_5C_2 = 1/10$. P(E among 3 chosen of 5) = ${}_4C_2/{}_5C_3 = 6/10 = .6$. Excel: =COMBIN(n , r), =PERMUT(n , r), =FACT(n).

§10 Random Variables — PMF & CDF Tables

Class 3

A random variable (RV) is a number decided by chance (X , Y , Z). Discrete = countable list (dice sum); continuous = measured (time, weight).

PMF / PDF — PROBABILITY OF EACH VALUE

$$p(x) = P(X=x) \cdot \text{needs } 0 \leq p(x) \leq 1 \text{ and } \sum p(x) = 1$$

CDF — RUNNING TOTAL, " $\leq$ "

$$F(x) = P(X \leq x) = \text{add } p \text{ up to } x \cdot P(X > x) = 1 - F(x) \cdot \text{reverse: } p(x) = F(x) - F(\text{prev})$$

WORKED TABLE — TOSS A COIN 3x, X = # HEADS

x	0	1	2	3
p(x)	1/8	3/8	3/8	1/8
F(x)	1/8	4/8	7/8	8/8

$p(2) = P(\text{exactly } 2) = 3/8$. $P(X \leq 2) = F(2) = 7/8$. $P(X > 2) = 1 - 7/8 = 1/8$. **Reverse:** if $F(10) = .3$, $F(20) = .9$, $F(50) = 1 \rightarrow p(20) = .9 - .3 = .6$.

§11 Expected Value, Variance & SD of a Discrete RV

Class 3

MEAN / EXPECTED VALUE

$$\mu = E(X) = \sum x \cdot p(x) \quad (\text{each value} \times \text{its probability, summed})$$

VARIANCE — USE THE SHORTCUT

$$\text{Var}(X) = \sum x^2 \cdot p(x) - \mu^2 = E(X^2) - [E(X)]^2 \cdot \text{SD} = \sqrt{\text{Var}}$$

RECIPE · E & VAR FROM A TABLE

1. $E(X)$ = add up $x \cdot p(x)$.
2. $E(X^2)$ = add up $x^2 \cdot p(x)$.
3. $\text{Var} = E(X^2) - [E(X)]^2$. $\text{SD} = \sqrt{\text{Var}}$.

NOTE The probabilities must sum to 1 before you trust E/Var. "Fair value of a gamble / game" = $E(X)$.

EXAMPLE — COIN, X = # HEADS (FROM §10)

$E(X) = 0(\frac{1}{8}) + 1(\frac{3}{8}) + 2(\frac{3}{8}) + 3(\frac{1}{8}) = 12/8 = 1.5$. $E(X^2) = 0 + \frac{3}{8} + 12/8 + 9/8 = 24/8 = 3$. $\text{Var} = 3 - 1.5^2 = .75$; $\text{SD} = \sqrt{.75} = .87$.

§12 Full Worked PDF — Distribution of a Sum $Z = X + Y$

Sample-Exam style

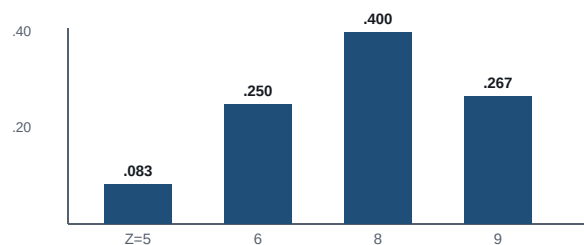
The game: roll a die. 5–6 → profit $X = \$2$ & draw Jar 1 (15R, 5G); 1–4 → profit $X = \$5$ & draw Jar 2 (8R, 12G). Ball profit Y : red = \$4, green = \$3. $P(5-6) = \frac{1}{3}$, $P(1-4) = \frac{2}{3}$. Build $P(Z)$ by multiplying each path, then plot.

RECIPE · BUILD A DISTRIBUTION

1. List every path; P = product along the path.
2. $Z = X + Y$ for each; merge equal Z 's.
3. Plot $P(Z)$ vs $Z \rightarrow$ that's the PDF.

THE FOUR PATHS

Case	X	Y	Z	P(Z)
5–6 & Green	2	3	5	$\frac{1}{3} \cdot \frac{1}{4} = .0833$
5–6 & Red	2	4	6	$\frac{1}{3} \cdot \frac{3}{4} = .2500$
1–4 & Green	5	3	8	$\frac{2}{3} \cdot .6 = .4000$
1–4 & Red	5	4	9	$\frac{2}{3} \cdot .4 = .2667$
Total				1.0000



SQUEEZE NUMBERS OUT OF THE PDF

$$E(Z) = \sum z \cdot P = 5(.083) + 6(.25) + 8(.40) + 9(.267) = 7.52$$

$$E(Z^2) = \sum z^2 \cdot P = 25(.083) + 36(.25) + 64(.40) + 81(.267) = 58.28 \rightarrow \text{Var}(Z) = 58.28 - 7.52^2 = 1.78, \sigma_z = \sqrt{1.78} = 1.34.$$

$$E(XY) (\text{product}) = \sum (x \cdot y) \cdot P = 6(.083) + 8(.25) + 15(.40) + 20(.267) = 13.83. \text{ This feeds covariance in §13.}$$

§13 Rules for E & Var · Covariance · Correlation

Class 3

EXPECTED VALUE ALWAYS ADDS

$$E(X+Y)=E(X)+E(Y) \text{ (true even if dependent)} \cdot E(aX+b)=a \cdot E(X)+b$$

VARIANCE OF A SCALED/SHIFTED RV

$$\text{Var}(aX+b) = a^2 \cdot \text{Var}(X) \text{ (+b vanishes; a gets squared)} \cdot \text{SD}(aX)=|a| \cdot \text{SD}(X)$$

ADDING RVS — INDEPENDENT

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

ADDING RVS — DEPENDENT (GENERAL)

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2 \cdot \text{Cov}(X,Y)$$

#1 EXAM TRAP VARIANCES add, SDs do NOT — never add standard deviations. Watch the sign: $\text{Var}(X-Y)$ still $+\text{Var}(Y)$ but -2Cov .

COVARIANCE

$$\text{Cov}(X,Y)=E(XY)-E(X) \cdot E(Y) \cdot \text{independent} \Rightarrow \text{Cov}=0$$

CORRELATION (UNITLESS, $-1 \dots 1$)

$$\rho = \text{Cov}(X,Y)/(\sigma_X \cdot \sigma_Y) \Rightarrow \text{Cov} = \rho \cdot \sigma_X \cdot \sigma_Y$$

SCALING

$$\text{Cov}(aX,bY)=ab \cdot \text{Cov}(X,Y) \cdot \text{Var}(aX+bY)=a^2\text{Var}(X)+b^2\text{Var}(Y)+2ab \cdot \text{Cov}(X,Y)$$

RECIPE · VAR OF $AX+BY$

1. Square each coefficient onto its own variance: $a^2\text{Var}(X)+b^2\text{Var}(Y)$.
2. Add cross term $2ab \cdot \text{Cov}(X,Y)$.
3. Independent? $\text{Cov}=0 \rightarrow$ drop the cross term.

WORKED COVARIANCE SET ($\Sigma_X=10, \Sigma_Y=12, \Sigma_Z=20$; $P_{XY}=-.8, P_{XZ}=-.5, P_{YZ}=.6$)

$$\text{Cov}(X,Y)=.8 \cdot 10 \cdot 12=\mathbf{96}; \text{Cov}(X,Z)=-.5 \cdot 10 \cdot 20=\mathbf{-100}; \text{Cov}(Y,Z)=.6 \cdot 12 \cdot 20=\mathbf{144};$$

$$\text{Cov}(2X,3Y)=6 \cdot 96=\mathbf{576}.$$

$$\text{Var}(X+Y)=100+144+2(96)=\mathbf{436}. \text{Var}(3X+5Z)=9(100)+25(400)+2 \cdot 15(-100)=\mathbf{7900}.$$

EXAMPLE — PORTFOLIO OF 2 ASSETS (HW2)

Weights $w_1=.4, w_2=.6$. $E=.12/.19, \sigma=.09/.28, \text{corr}=-.4$. Return $R=.4r_1+.6r_2$.

$$E(R)=.4(.12)+.6(.19)=\mathbf{.162}. \text{Cov}=-.4(.09)(.28)=-.01008. \text{Var}(R)=.4^2(.09^2)+.6^2(.28^2)+2(.4)(.6)(-.01008)=\mathbf{.02468}; \sigma(R)=\mathbf{.157}. \text{Weights are the } a,b \text{ — same } aX+bY \text{ rule.}$$

§14 The Named Discrete Distributions

Class 3 / HW

Binomial — "how many successes in n independent tries?"

Spot it: *FIXED* n trials · only 2 outcomes (success/fail) · same π each time · trials independent (WITH replacement).

$$P(X=k)={}_nC_k \cdot \pi^k \cdot (1-\pi)^{n-k} \cdot E(X)=n\pi \cdot \text{Var}(X)=n\pi(1-\pi) \cdot \text{SD}=\sqrt{n\pi(1-\pi)}$$

"At least one" trick: $P(X \geq 1)=1-(1-\pi)^n$. Excel =BINOM.DIST(k,n, π ,FALSE/TRUE).

BINOMIAL EXAMPLES

Salesperson $n=25, \pi=.3$: $p(8)={}_{25}C_8 \cdot .3^8 \cdot .7^{17}=\mathbf{.165}$; $E=7.5$; $\text{SD}=\sqrt{25 \cdot .3 \cdot .7}=\mathbf{2.29}$. **Sample bonus** $\pi=.267, n=5$: $P(\geq 1)=1-.733^5=\mathbf{.788}$; $P(\text{exactly } 2)={}_5C_2 \cdot .267^2 \cdot .733^3=\mathbf{.280}$.

OVERBOOKING — EXPECTED VALUE OF A BINOMIAL FUNCTION (HW2)

40 rooms, 50 reservations, $P(\text{show})=.8 \rightarrow X \sim \text{Bin}(50,.8)$; pay \$150 per guest over 40. $E[\text{comp}]=150 \cdot \Sigma_{x=41..50} (x-40)P(X=x)=150(1.119)=\mathbf{\$167.78}$.

Poisson — "how many random events in a fixed span?"

Spot it: events at a steady rate over time/space, no upper limit on the count.

$$P(X=k)=e^{-\lambda} \cdot \lambda^k/k! \cdot E(X)=\text{Var}(X)=\lambda \cdot (\text{SD}=\sqrt{\lambda}) \cdot \lambda = \text{rate} \times \text{window}$$

MATCH THE WINDOW Rescale λ to the interval asked: 4/hr $\rightarrow \lambda=8$ for 2 hours, $\lambda=1$ for 15 min. "At least one" $=1-e^{-\lambda}$.

POISSON EXAMPLES (CALL CENTER, 4/HR)

$P(5 \text{ in } 2 \text{ hr})$: $\lambda=8 \rightarrow e^{-8}8^5/5!=\mathbf{.0916}$. $P(0 \text{ in } 15 \text{ min})$: $\lambda=1 \rightarrow e^{-1}=\mathbf{.368}$. $P(\geq 1 \text{ in } 1 \text{ hr})$: $\lambda=4 \rightarrow 1-e^{-4}=\mathbf{.982}$. $P(\geq 3)$, $\lambda=2$: $1-e^{-2}(1+2+2)=\mathbf{.323}$.

Hypergeometric — "without replacement" (chances change each draw)

$$P(X=k)=[C(K,k) \cdot C(N-K, n-k)] / C(N,n)$$

N =population, K =successes in it, n =draws, k =successes wanted. **With replacement** – Binomial; **without** – Hypergeometric.

HYPERGEOMETRIC EXAMPLES

Marbles 20 (12R), draw 5: $P(3 \text{ red})=C(12,3) \cdot C(8,2)/C(20,5)=\mathbf{.397}$. **Acceptance sampling** (HW2): ship $N=200$, inspect 12, reject if any of 10 defectives appears: $P(\text{reject})=1-C(190,12)/C(200,12)=\mathbf{.469}$ (binomial approx $1-.95^{12}=.460$).

§15 The Normal Distribution & the Z-Score

Class 4

Bell-shaped, symmetric about μ ; two parameters μ (mean) and σ^2 (variance). Continuous $\Rightarrow P(X=a)=0$, so "<" and " $\leq$ " are interchangeable. Uniform aside: on $[1,5]$ density height $=\frac{1}{4} \rightarrow P(2 \leq X \leq 4)=\frac{1}{2}$.

STANDARDIZE — THE MASTER MOVE

$$Z = (X - \mu)/\sigma \rightarrow \text{look Z up in the exam's normal table } (Z \sim N(0,1))$$

RECIPE · NORMAL PROBABILITY

1. Turn each X into a Z: $(X-\mu)/\sigma$.
2. Look up $P(Z \leq z)$ in the table.
3. "Greater" $\rightarrow 1 -$ table. "Between" $\rightarrow$ subtract the two table values.

GO BACKWARDS — FIND X FOR A GIVEN %

$X = \mu + z \cdot \sigma$ (look up the z for the probability first; z is – below the mean)

EXAMPLE — COKE DEMAND $N(M=150, \Sigma=20)$

$$P(X \leq 200): z=2.5 \rightarrow \mathbf{.9938}. P(X \geq 180): z=1.5 \rightarrow 1-.9332=\mathbf{.0668}.$$

$$P(150 \leq X \leq 180)=.9332-.5=\mathbf{.4332}. \text{Stock so } P(\text{run out})=.10 \rightarrow \text{top } 10\% \rightarrow z=1.28 \rightarrow X=150+1.28(20)=\mathbf{176 \text{ cans}}.$$

§16 Central Limit Theorem — Sums & Averages

Class 4

The CLT: add up (or average) many independent, same-distribution RVs and the result is ≈ Normal — even if each piece is NOT normal. Condition: independent + identically distributed + n large (n ≥ 30). If the pieces are already Normal, it's EXACT for any n.

SUM OF N	AVERAGE OF N ($\bar{X}$)
mean = $n\mu$	mean = μ
var = $n\sigma^2$	var = σ^2/n
SD = $\sigma\sqrt{n}$	SE = $\sigma/\sqrt{n}$
$Z = (T - n\mu)/(\sigma\sqrt{n})$	$Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$

SUM VS AVERAGE SD of a **sum** = $\sigma\sqrt{n}$ (grows); SD of an **average** = $\sigma/\sqrt{n}$ (shrinks). Standardize with the matching SD. And "n>30 makes an individual value normal" is **FALSE** — CLT only normalizes the sum/average.

SUM — CARS SOLD
E(X)=1.7, Var=.81. Over 300 days:
mean=510, SD= $\sqrt{(300 \cdot .81)}$ =15.59.
 $P(<495)=P(Z<-.96)=$ **.168**.

AVERAGE — BRANCH EMPLOYEES
 $\mu=50, \sigma=15, n=40$: SE=2.37 →
 $P(\bar{X}>55)=P(Z>2.11)=$ **.017**. n=100:
SE=1.5 → z=3.33 → **.0004**.

CORN CHIPS (SAMPLE EXAM Q10): M=11.95, Σ =.22, N=100
(a) $P(\text{sum}>1194)$: mean=1195, SD= $\sqrt{100 \cdot .22}$ =2.2, $z=(1194-1195)/2.2=-.45$ → $P(>)=$ **.675**
(b) $P(\bar{x}<11.97)$: SE=.022, z=.909 → **.818**. (c) Is one bag Normal because n>30? **No**
— CLT normalizes the average/sum, never a single item.

NORMAL ≈ BINOMIAL If $n\pi>5$ and $n(1-\pi)>5$, treat Binomial X as Normal with mean= $n\pi$, var= $n\pi(1-\pi)$. Ex: insurance n=1000, $\pi=.06$ → mean=60, SD=7.51;
 $P(X<70)=P(Z<1.33)=$ **.908**.

§17 The Normal Table — Reading & Interpolating

Class 4

The exam's table gives $P(Z \leq z)$ = area to the **LEFT** of z. First turn any question into a Z: $z = (X - \mu)/\sigma$. The only 4 moves: ① $P(Z \leq z)$ =read it straight off · ② $P(Z > z) = 1 - P(Z \leq z)$ · ③ $P(Z \leq -z) = 1 - P(Z \leq z)$ (symmetry) · ④ $P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$.

TWO TABLE STYLES — CHECK YOURS Some tables instead list $P(0 \leq Z \leq z)$ (area from the middle out). Then $P(Z \leq z) = 0.5 + \text{that}$. Ex: a middle-table 1.35 shows .4115, so $P(Z \leq 1.35) = .5 + .4115 =$ **.9115**. Same answer — just add the 0.5.

INTERPOLATE FORWARD (HAVE Z, WANT AREA)
 $P(z) = P(z_1) + \frac{(z - z_1)}{(z_2 - z_1)} \cdot [P(z_2) - P(z_1)]$

INTERPOLATE BACKWARD (HAVE AREA, WANT Z)
 $z = z_1 + \frac{(A - A_1)}{(A_2 - A_1)} \cdot (z_2 - z_1)$

Fwd ex: $P(Z \leq 1.234)$: $P(1.23) = .8907, P(1.24) = .8925$ → $.8907 + .4(.0018) =$ **.8914**. **Back ex:** area .90 sits between 1.28(.8997) & 1.29(.9015) → $1.28 + .17(.01) =$ **1.282**.

Common critical values — memorize the bold ones

z	P(Z≤z)	area right	use
1.00	.8413	.1587	$\mu + \sigma$
1.28	.8997	.10	top 10%
1.645	.9500	.05	top 5%
1.96	.9750	.025	top 2.5%
2.33	.9901	.01	top 1%
2.575	.9950	.005	top 0.5%

§18 "Which Formula?" — the Decision Guide

If the problem says...	Use this
"OR", "either", "at least one of A/B"	Addition: $P(A) + P(B) - P(A \cap B)$
"AND", "both"	Multiplication: $P(A B) \cdot P(B)$; independent → $P(A) \cdot P(B)$
"neither" / "only A, not B"	$1 - P(A \cup B)$ / $P(A) - P(A \cap B)$
"given", "if... then chance of..."	Conditional: $P(A B) = P(A \cap B)/P(B)$
"reverse the given" (know effect, want cause)	Bayes / Total Probability
"how many ways", order matters (ranks)	Permutation ${}_nP_r$
"how many ways", pick a group (order NO)	Combination ${}_nC_r$
fixed # of yes/no trials, "exactly k / at least"	Binomial ${}_nC_k \pi^k (1-\pi)^{n-k}$; " ≥ 1 " = $1 - (1-\pi)^n$
events over time/space at a rate, no upper limit	Poisson $e^{-\lambda} \lambda^x / x!$, λ =rate-time
draw without replacing	Hypergeometric
"normally distributed", one item, $P(X \leq \dots)$	$Z = (X - \mu)/\sigma$, then table
"total / sum of n things"	CLT sum: mean $n\mu$, SD $\sigma\sqrt{n}$
"average / mean of n things"	CLT average: mean μ , SD $\sigma/\sqrt{n}$
find the value for a target % (e.g. top 10%)	$X = \mu + z \cdot \sigma$ (look up z first)

FINAL 3 REMINDERS (1) Sample variance uses **n-1**. (2) Variances add, SDs never do. (3) CLT normalizes the **sum/average**, not a single observation.

§19 The Sample Midterm — Every Problem in One Table

#	Problem	What it tests	Method	Answer
1	P(select a green ball)	Total probability	Σ path products: $.25(\frac{1}{3})+.6(\frac{2}{3})$	.4833
2	P(rolled 5–6 green)	Bayes	one path $\div$ P(green): $.0833/.4833$	.1724
3	P(rolled 1–4 $\cup$ green)	Addition rule	$P(A)+P(B)-P(A\cap B)$	.75
4	Distribution of profit $Z=X+Y$	Build a PMF	list 4 paths, multiply, merge (\$12)	Z=5,6,8,9
5	$E(Z)$ and σ_Z	Mean & SD of an RV	$\Sigma z \cdot P; \sqrt{(\Sigma z^2 P - \mu^2)}$	7.52 · 1.34
6	$E(XY)$	Expected product	$\Sigma(x \cdot y) \cdot P$ over the 4 cases	13.83
7	Win \$9 at least 25 $\times$ in 100	Binomial $\approx$ Normal	$np=26.67$, $\sigma=4.42$, standardize	$\approx .69$
8	$\text{Var}(2X)$ vs $\text{Var}(X+Y)$	Variance rules	$2^2(49)=196$ vs $49+49=98$	196 > 98
9	Are A, B independent?	Independence test	does $P(A)P(B)$ match the given?	No
10a	P(total of 100 bags > 1194)	CLT — sum	mean $n\mu=1195$, SD $\sigma\sqrt{n}=2.2$	.675
10b	P(sample mean < 11.97)	CLT — average	$SE=\sigma/\sqrt{n}=.022$, $z=.909$	.818
10c	Is one bag $\approx$ Normal ($n>30$)?	CLT concept	applies to mean/sum, not an item	No